



Advancement in medical report generation: current practices, challenges, and future directions

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Abstract

The correct analysis of medical images requires the medical knowledge and expertise of radiologists to understand, clarify, and explain complex patterns and diagnose diseases. After analyzing, radiologists write detailed and well-structured reports that contribute to the precise and timely diagnosis of patients. However, manually writing reports is often expensive and time-consuming, and it is difficult for radiologists to analyze medical images, particularly images with multiple views and perceptions. It is challenging to accurately diagnose diseases, and many methods are proposed to help radiologists, both traditional and deep learning-based. Automatic report generation is widely used to tackle this issue as it streamlines the process and lessens the burden of manual labeling of images. This paper introduces a systematic literature review with a focus on analyses and evaluating existing research on medical report generation. This SLR follows a proper protocol for the planning, reviewing, and reporting of the results. This review recognizes that the most commonly used deep learning models are encoder-decoder frameworks (45 articles), which provide an accuracy of around 92–95%. Transformers-based models (20 articles) are the second most established method and achieve an accuracy of around 91%. The remaining articles explored in this SLR are attention mechanisms (10), RNN-LSTM (10), Large language models (LLM-10), and graph-based methods (4) with promising results. However, these methods also face certain limitations such as overfitting, risk of bias, and high data dependency that impact their performance. The review not only highlights the strengths and challenges of these methods but also suggests ways to handle them in the future to increase the accuracy and timely generation of medical reports. The goal of this review is to direct radiologists toward methods that lessen their workload and provide precise medical diagnoses.

Keywords Automated medical health services · Deep learning · Image processing · Public health · Report generation

1 Introduction

The analysis of different medical images, such as chest X-rays, computed tomography (CT), magnetic resonance imaging (MRI), and other pathological images, requires detailed analysis by professional radiologists to determine the correct diagnosis. Once the disease is diagnosed, it is the responsibility of the radiologist to write a detailed and well-structured report. Medical reports are based on natural language generation (NLG) that generates textual scripts from different sources. This helps radiologists write medical reports for various diseases. However, analyzing and writing reports can be time-consuming and require manual effort.

To reduce the burden on radiologists, automated medical report generation plays a vital role. It uses key components of machine learning and deep learning to effectively analyze and automate the report writing procedure. Doing so leads to a more efficient process for report generation. The automation of medical reports indeed paves the way for increasing accuracy in report generation; however, the clinical relevance of the reports should be maintained.

Image captioning [1] is a process that automatically generates text descriptions. Studies [2–4] show the effectiveness of image captioning in generating reports similar to how radiologists write them by understanding the semantics. This process aims to streamline the entire workflow of report generation with reduced time and more efficient results, along with aiding radiologists. It uses an encoder-decoder framework [5], where convolutional neural networks (CNN)

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extract visual information from images and the decoder, a long short-term memory (LSTM) model with natural language processing (NLP), extracts and generates textual descriptions, highlighting important information. However, report generation is a complex process, and it is crucial to understand the details in depth. Other variations of image captioning with neural networks, such as recurrent neural networks (RNN) [6] that, when combined with CNN, showed effective results, are on a CNN-RNN encoder-decoder framework. Both these approaches require further research and development to provide more efficient results. Transformer architecture also provides a robust solution to the generation of accurate medical reports as they interpret complex data and understand vital patterns.

Medical images are complex, and it is difficult to understand the underlying patterns. If images have multiple views, it becomes even more difficult to understand them. Simple image captioning often struggles to capture in-depth details. Extensive research has been done for generating automated medical reports. The study [7] uses a transformer-based architecture to generate reports. Transformers can further be fine-tuned on inputs such as visual and semantic features capturing both for diagnostic report generation. Similarly, [8] handle multi-view images using encoder-decoder frameworks to assist in the identification of complex patterns. Recurrent neural networks [9] and attention mechanisms integrated with CNN and LSTMS [10] are more precise ways to automate the report generation process leading to more accurate reports.

Various methods provide promising solutions to report generation of medical images to enhance accuracy and consistency. However, there is still room for improvement. Current methods often face challenges in capturing complex medical details as discussed in [11, 12]. Another issue is the quality of data and bias in the training data [2, 13]. Overall, all these hurdles potentially arise due to the need to extract more discriminate features to understand the complex patterns of the images. As existing methods require pixel-level annotated images, it can sometimes be time-consuming and can cause inaccuracy.

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1.1 Existing surveys

Automatic medical report generation is gaining immense popularity all over the globe, and a substantial amount of work is done in this regard. Mostly, deep learning methods are used to tackle the research gaps associated with the report generation of medical images. The study [14] presents a survey on the advancements of deep learning for report generation. The survey covers image processing-based captioning methods. This forms the basis for generating text from images. The authors discuss several deep learning methods such as encoder-decoder frameworks, hierarchical RNN-based frameworks, attention-based frameworks, and reinforcement learning-based frameworks. Potential challenges related to these methods are explained along with suggestions for further development.

Deep learning methods have made the progress in generating textual reports quite easy, as discussed in [15]. The most commonly used method is CNN, which is used in almost all sorts of report generation for all kinds of images. The study [15] also discusses RNNs and a combination of both known as the encoder-decoder framework.

As image captioning is the foundation of generating text from images, the study [16] discusses its importance. This survey focuses on methodologies like computer vision, NLP, and paod review protocol. These methods provide a robust foundation in the generation of medical reports and also provide ways to improve the accuracy further. Similarly, the survey [17] focuses on deep learning methods such as encoder-decoder frameworks, attention-based frameworks, retrieval-based methods, and knowledge graphs for the generation of medical reports. Table 1 describes existing surveys along with their contribution and shortcomings.

1.2 Contributions of current study

This survey focuses on a broader aspect of medical report generation, while existing surveys focus on those methods that involve deep learning solely. However, this survey touches on all aspects of medical report generation.

- It focuses on those methods as well that do not involve any deep learning and focuses on traditional methods as well that the radiologists follow.
- It also includes a broader range of deep learning methodologies along with CNNs and RNNs such as encoder-decoder frameworks, transformer-based frameworks, attention-based frameworks, large language models, and graph-based frameworks.

Table 1 Summary of surveys with their respective image modalities, methodologies, drawbacks, and observations

Study	Image modality	Methodology	Drawback	Observations
[14]	Chest X-rays	Encoder-decoder frameworks, hierarchical RNN-based framework, attention-based framework, reinforcement learning-based framework	Limited data diversity and generalizability	Promising initial results, needs more diverse datasets
[18]	Chest X-rays, MRI, PET	CNNs, RNNs, encoder-decoder framework	Integration challenges of CNN and RNN, and scalability issues	Significant progress in combining image analysis and NLP
[16]	Chest X-rays	Computer vision, NLP, PRISMA review protocol	Complexity in accurately linking image features to clinical findings	Strong foundation for improving medical report accuracy
[17]	Chest X-rays	Encoder-decoder frameworks, attention-based frameworks, retrieval-based methods, knowledge graphs	High computational cost, and need for large annotated datasets	Advanced techniques integrating clinical knowledge show promise

- This survey also focuses on different image modalities such as MRIS, CTs, and X-rays instead of focusing on just one.

It follows a proper review paper protocol and proper quality assessment method to analyze and understand various studies on this topic. The aim is to provide a good source for further research and development. Having a more in-depth review can further guide researchers to carry out advancements in automated medical report generation.

Traditional medical report generation methods [19] and automated methods are both important, but two factors determine the correctness of reports: accuracy and clinical relevance. The non-machine learning methods rely on the radiologist's knowledge to analyze the images and then diagnose the disease. The automated methods take help from machine learning and deep learning methods for diagnosis and the fast generation of medical reports. This survey has the following objectives:

- Identify knowledge gaps:* Identify and address gaps in existing research on medical report generation methods.
- Improve the clinical process:* Helps to streamline the report generation process so as to reduce the radiologists' burden.
- Enhance model accuracy:* Evaluate and propose changes to enhance accuracy in deep learning models for medical report generation.
- Facilitate future research:* Offers a detailed review that will inform directions for future research efforts on addressing current limitations and challenges.
- Support clinical decision-making:* Improve the reliability and timeliness of medical reports, thereby aiding clinical decision-making and patient care.

2 Literature review

2.1 Non-machine learning methods

Traditional methods for report generation rely on the expertise of radiologists to analyze and diagnose diseases and then write detailed reports. The traditional way involves intrinsic and extrinsic methods as in the paper [19]. To assess the coherency of the text, it uses techniques such as Likert scale scoring or pairwise comparisons and it helps ensure good quality of the text but can be time-consuming and expensive. Extrinsic methods focus more on user-centric metrics but can have high variability. Both these methods help the radiologist write their reports.

Another study [20] also discusses traditional methods for report generation. It focuses on the textual aspect and what contributes to that. In this survey, descriptive statistics of the data are carried out after the images are analyzed; this helps in making predictions and identifying diseases. It also performs a qualitative analysis and usability testing to understand the radiologist's perspective and also focuses on increasing the accuracy of medical reports without ML methods in the traditional way. Other methods are simple; the radiologists analyze the images, and based on their knowledge and assessment, they write the textual report that is the most traditional way.

Structural reporting [21] is also a way to write medical reports. This paper explains the importance of structural reporting in contrast to free text reporting. Free text reporting maintains the individuality of radiologists, meaning they can explain what they think is right, and this can lead to a lot of discrepancies in the results, impacting the results. Structural reporting helps to solve this issue by utilizing standardized checklists which ensure that all necessary items and details are covered and not skipped. It is made sure that this checklist

is then communicated with the physicians as well for a better understanding. This provides a more robust and uniform approach minimizing bias and inaccuracy in the results and helps to generate more accurate medical reports.

All these methods indeed provide good ways for writing medical reports; however, they can increase the cost and can be quite time-consuming. This can cause a burden on the radiologists and may lead to delay and inaccurate reports, so there is room for further improvement.

2.2 Encoder-decoder frameworks

The authors of the study [11] offer an adaptable approach that includes gathering data, choosing a model, and analyzing the findings. The IU-Xray dataset provides XML reports and X-ray images. After preprocessing, a structured dataset is produced, from which a trained model named Chexnet extracts pertinent features. Sequential data processing is carried out utilizing an encoder-decoder system that uses LSTM and GRU algorithms to overcome long-term dependency. Because of its pre-trained model, this method not only reduces the amount of manual annotation but also generates reports with higher accuracy. However, some problems arise, like the potential for bias, which results in an overabundance of training data and computational resources that need to be decreased in order to achieve better results.

Another study using encoder-decoder frameworks has demonstrated effective outcomes in recent studies for the creation of medical reports [10, 11, 22]. ConvNeXt as an encoder and BioBERT as a decoder together offer a beneficial development for the production of medical reports [23]. ConvNeXt's hybrid CNN architecture helps to better comprehend the location of lesions by extracting significant information from the image. It is limited, nevertheless, in that it cannot capture the image's long-range dependencies. The transformer-based decoder BioBERT is excellent in comprehending language in many contexts and adapting to particular domains; yet, it has problems because of its computational resources and its somewhat opaque decision-making process. Despite these drawbacks, this concept offers a possible remedy for accurate report generation.

A related model [24] for medical report generation utilizes deep learning of COVID-19 from CT scans. Medical-VLBERT, a combination of transfer learning and encoder-decoder framework, is proposed by the authors to handle the issue. It is first pretreated on the CX-CHR data set and then fine-tuned on the COVID-19 dataset to diagnose diseases. For the encoder-decoder, it utilizes DenseNet-121 for feature extraction and VLBERT for the textual terminology. This approach streamlines the process of medical report generation by gathering extensive information. These methods show promising solutions to handle report generation and appear to be efficient.

Encoder-decoder frameworks are the most used method for the generation of medical reports. They effectively capture and extract information for all types of image modalities. These frameworks focus on generating accurate reports using encoders, mostly CNNs and decoders of different kinds. Similar approaches are mentioned in the papers [25–30] which focus on these frameworks to produce accurate reports.

2.3 Transformer-based frameworks

There are two primary methodologies used in this research [6] for the generation of radiological reports from the CXR images. The first one is the pre-processing and feature extraction, and the second one is the integration of hierarchical LSTM (HLSTM) with pre-trained BERT embeddings. Pre-processing of the data is done, and visual features are extracted by the VGG19 network with the assistance of pre-processing and feature extraction methodologies. This way, the data is made comparable and handy. Well-formatted input data with relevant features assures the generation of more accurate reports, which is in turn facilitated by the HLSTM and pre-trained BERT embedding approach. This is achieved by encoding both local and global dependencies in the textual content and finally yields contextually meaningful reports. But there are some drawbacks, though. Some of them are computational complexity, fine-tuning requirements, and overfitting capacity. Despite all these drawbacks, the performance of deep learning techniques in generating radiological reports from the CXR images is good.

The study [11] focuses on using transformer-based architectures to generate reports for chest X-rays. The model put forward in the research effectively handles complicated interactions in medical data by capturing contextual subtleties and long-term reliability for consistent medical report production. Large finished data and parameters are handled efficiently while employing the DenseNet-121 CNN architecture for feature extraction. Moreover, there are still certain issues with how well it captures the intricacy of the visuals. Furthermore, analytical methods that depend on common metrics for quantitative evaluation, such as BLEU, METEOR, and ROUGE, may also miss important details in reports that are produced.

The authors in [22] introduce a CNN DenseNet121 which is good at extracting different features from chest X-ray images and re-using them. The need to adjust and train images to extract relevant features is still a challenge it possesses. To generate the reports from the features, a GRU decoder understands the sequential data and how it relates to other text, but it struggles with understanding complex medical terms. Dropout techniques are used to reduce the chances of overfitting, but dealing with the best dropout rate and computational resources can be a challenge. However, combining these approaches provides significant improve-

ment in generating medical reports if all limitations are kept in mind and dealt with accordingly.

An approach called adaptive image-tag-merging network (AIMNet) (presented by the researchers [23]) demonstrates the generation of accurate reports by combining visual features from medical images with labels of the diseases. It utilizes advanced attention mechanisms such as visual and tag-attention to combine the important details; the adapting merging gate adjusts how much focus is on images and words, making detailed reports according to each patient's situation. AIMNet is built on a transformer-based architecture, which makes it easy to adjust and expand vast medical information; however, it is complex and biases from the training data can cause certain hurdles. Despite these limitations, it is still a good approach for generating medical reports.

The studies [31–39] utilize transformed-based models for the generation of medical reports. These approaches increase the accuracy and correctness of report generation. As transformers are capable of capturing large dependencies in data, it makes them useful for this purpose. However, they do face limitations as they can be computationally expensive. Overall, approaches using transformers offer promising solutions.

2.4 RNN-LSTM-based frameworks

Semantic fusion network (SFNet), a two-module process, uses lesions area detection module and report generation as presented in the study [12]. The lesion area detection module utilizes a faster R-CNN to detect where lesions occur in the images. Visual and disease-relevant information are combined using LSTM to generate reports. Word2vec with a sentinel gate mechanism in LSTM is used to generate grammatically correct sentences and contain precise disease information that potentially leads to early diagnoses and better accuracy. Despite providing advantages, this approach has its issues, that is, potential inaccuracy in lesion detection due to bias in the training data and limited ability to handle complex patterns of data. Overall, this approach provides a way to enhance medical report generation.

Deep learning strategies are being actively explored to generate automated medical reports from chest X-ray (CXR) images [40]. The objective is to streamline diagnosis and reduce the burden on radiologists. This paper proposes certain techniques that produce good results for generating reports. Encoder-decoder models with attention mechanisms, sequence-to-sequence transformers, and LSTM, GRU, and Bi-GRU algorithms provide improved report quality, accuracy, and workflow integration. However, computational complexity and understandability of data are certain challenges that occur. Focus on solving these issues with the use of better methods leads to a better generation of medical reports.

An encoder-decoder framework [41], where CNNs as the encoder and LSTM as the decoder generate reports from images, streamlines the workflow and reduces the burden on radiologists. Transfer learning techniques such as inception-v3 utilize pre-trained CNNs and word embedding such as GloVe showing a better understanding of larger datasets, especially limited annotation data. RadGLove embedding shows an improved understanding of complex medical terminology that enhances report generation. These approaches have their limitations, which are dealing with long reports and addressing bias in embeddings which resulted in variations in the reports.

To understand the complexity of radiographs for report generation, the framework proposed by the authors in [42] utilizes the encoder ResNet-152 to extract visual features from the X-ray images and a transformer decoder using memory feature affine transformation (MFAT) to generate text reports and improve the flow of text. Along with the framework, the authors propose a novel system to handle class imbalance known as the imbalance-aware clinically accurate reward (ICAR), which increases the framework's ability to identify any correct or incorrect data. However, the MFAT may increase time in training, which requires further enhancement of the approach.

The authors of the study [13] introduce a new concept flexible multi-view paradigm (FMVP) that analyzes the image and predicts diseases similar to that of a radiologist. The purpose is to ease the burden on radiologists and enhance the accuracy of medical report generation. The two methods for this concept are case-based retrieval (CBR), which finds similar cases in the training and generates reports, and disease-based retrieval (DBR), which utilizes a pre-defined library to predict diseases. FMVP shows good results on the NLG metric, but it still requires proper clinical validation. However, due to this, it is easier to analyze images and generate reports.

RNNs can capture patterns in sequential data, therefore improving the report generation process. LSTMs are good at generating long sentences, which are crucial for detailed reports. Papers [43–46] focus on approaches using RNNs. These approaches are paired with CNN, which extracts the information from the images, and then RNNs generate reports. Even though these models handle large dependencies really well, they often struggle with training large numbers of data and lead to overfitting.

2.5 Attention-based frameworks

Medical image analysis is a progressing field with innovative approaches to generating accurate medical reports. An encoder-decoder framework that extracts both visual and semantic features with the use of ResNet-152 and VGG-16, which are essential to generate clinically relevant reports

and a hierarchical LSTM with multi-attention mechanisms capturing the complexity of the data with the addition of Adversarial reinforcement learning, which is an iterative process that enhances the reports while still keeping them easy to understand and providing an accurate diagnosis. This is a solid approach as this paper mentions [47]. Despite this, it does face some drawbacks, such as dealing with the complexity, the dependency on high-quality annotations of the data, and sometimes sensitivity to the hyperparameters. Overall, this approach is a good step in enhancing medical report generation.

Attention-enhanced network with semantic inspector, a novel method proposed in [48], aims to capture and identify important regions in images and generate medical reports. To focus on informative regions, it uses a double-weighted multi-head attention that assigns weights based on information. For report generation, it analyzes the image tags for diseases during training and applies a tags prediction module to determine the quality of the reports. This approach aims to enhance medical report generation, but it has a high dependence on the dataset.

The study [49] focuses on a hierarchical system with multiple attention mechanisms. This approach captures both spatial and channel information of the image to enhance the understanding of the text and provide clearer sentences. This increases the robustness of the textual reports. However, it does face certain limitations such as complexity due to the hierarchical system, which leads to more training time and can delay the process and also impact accuracy.

Multi-attention mechanism and background information are two concepts that are being abundantly used in the generation of medical reports. Along with this, an encoder-decoder framework is also used. The authors [50] use ResNet-152 as the encoder and add multi-attention mechanisms for extracting better features, and this helps in capturing more complex patterns in the images. Despite this, it does face challenges in complexity and high computational resources.

The authors in this paper propose various ways to generate medical reports. They combine attention mechanisms with graph-based frameworks. Structured knowledge enhanced multi-model attention (SKEMA) is the framework proposed [51]. It utilizes RadGraph and PubMedBERT to gather relevant textual information from the images. It helps to integrate both visual and semantic information, making the generation of reports more accurate and timely. However, it depends on the quality of the knowledge graph, and this can impact its accuracy.

Attention mechanisms have made the generation of medical reports quite fast and accurate as well. Attention guided object dropout MLP (ODM) model is proposed in the paper [52]; it contains a pre-trained model that analyzes and extracts relevant features from the images that further help in the correct diagnosis of diseases. ODM also minimizes

the impact of irrelevant information, and only those features are selected that are important, but it faces challenges in generalizability to different datasets and can have complex computational resources.

2.6 Large language models

The authors of this paper [53] propose a method to generate reports using GLM and ChatGLM, which are large language models (LLMs). These methods help in generating more structured and coherent reports. LoRA and P-tuning, two parameter fine-tuning methods, are used for this purpose, and they help in reducing the computational costs and improving performance. However, it has its limitations and can be expensive to implement as a large number of data need to be present. Other than that, noisy data can also cause problems. Overall, this method provides promising results in medical report generation.

R2GenGPT is a framework that uses a visual encoder (swin transformer), a visual mapper, and llama2-7B (LLM) [54]. The combination of these transformer methods with the LLM model helps to provide a robust framework, and it increases the performance and efficiency by producing well-structured reports. However, it is computationally challenging, and fine-tuning on large datasets can lead to overfitting.

2.7 Graph-based frameworks

Three different parts consisting of chest X-rays, clinical data, and medical expertise are used to generate reports. The approach mentioned in the paper [55] uses a classifier, such as a graph convolutional network, to extract the features and a generator, such as an attention-based model, to generate the reports; all based on the information from the medical knowledge. This helps in producing more robust reports that have better accuracy; however, this approach is data-dependent, meaning the data's quality has an impact on the accuracy.

2.8 Multi-modal approaches

This novel approach [56] to automatic chest radiology report generation with the aid of multimodal feature fusion combines visual features from chest X-rays obtained using a Vision Transformer with semantic features from existing medical reports obtained by using Word2Vec. It improves the process of report generation over the disadvantages inherent in professionalism, disease variety, and fluency in the traditional encoder-decoder models that tend to treat the image-to-report relationship in a unidirectional way. Some benefits would be more consistent in medical images and reports; more coherent reports can be issued; results are

just better for those metrics such as ROUGE and BLEU. The potential downsides may be that the model relies on quality, labeled medical reports for training, and integration of advanced techniques, such as cross-modal information fusion, proves very complex and perhaps computationally resource-intensive, along with being sophisticated enough to require specialized knowledge and skills for practical application in clinical settings.

This new algorithm automatically generates radiology reports that would help overcome the major problems: limited and inefficient use of prior medical knowledge, knowledge fusion into reports, and either repetitive or indistinctive reporting [57]. It integrates SKEMA, a knowledge graph used for the fusion of medical knowledge of image features, and DBCL, which ensures distinctiveness between reports by comparing original and augmented features. It further enriches both image and knowledge-based features of positive and negative samples, averts the situation of the model to produce templated reports, and improves its robustness. Overall, the proposed approach showed superior performances on public datasets such as IU-Xray and MIMIC-CXR with higher evaluation scores compared to baseline models. The positives include a higher level of integration of clinical knowledge and more diversity in reporting, while some negatives may be the complexity of graph integration of knowledge and a greater need for computation.

In a medical set up, the most important thing is the privacy of the patient's data and information. Chen and Pan [58] focus on an approach known as multi-model federated learning, which is a combination of deep learning and distributed computing. This approach uses a federal average algorithm that merges models from different sources into one model. This not only optimizes the accuracy but also improves the generation of reports without communicating confidential information.

2.9 Explainable AI in medical report generation

Explainable AI (XAI) is an emerging phenomenon that explains decisions in a way similar to humans. It aides in the diagnosis of diseases easing the way for doctors as well as patients to trust and understand the reason behind certain medical decisions that help in providing a more precise diagnosis.

The paper [59] provides a review of how XAI can assist in the diagnosis of diseases and generate medical reports. It is an SLR that focuses on all possible applications and techniques that help in understanding AI-driven outcomes.

Another study discusses various XAI methods for a comprehensive understanding of AI-based diagnosis. Methods such as Layer-wise Relevance Propagation LRP, LIME, SHARP, and Grad-CAM are crucial for diagnosing tumors

and cysts as mentioned in [60]. It also explains the application of XGBoost for analyzing 3D brain tumor and mediastinal cysts. These methods indeed provide promising results, but they face challenges in effective implementation.

3 Methodology

When conducting a survey, it is important to follow a proper protocol to ensure correct and accurate reviewing of literature. Figure 1 shows the steps followed in this survey.

This survey follows the protocol outlined in [24], and it contains the following phases:

- i. Planning the process
- ii. Reviewing the process
- iii. Evaluation of results

3.1 Planning

The first phase of the survey is to plan the entire process and then carry out all the steps. In this phase, it is important to shortlist potential research questions. These are answered in subsequent sections of the study.

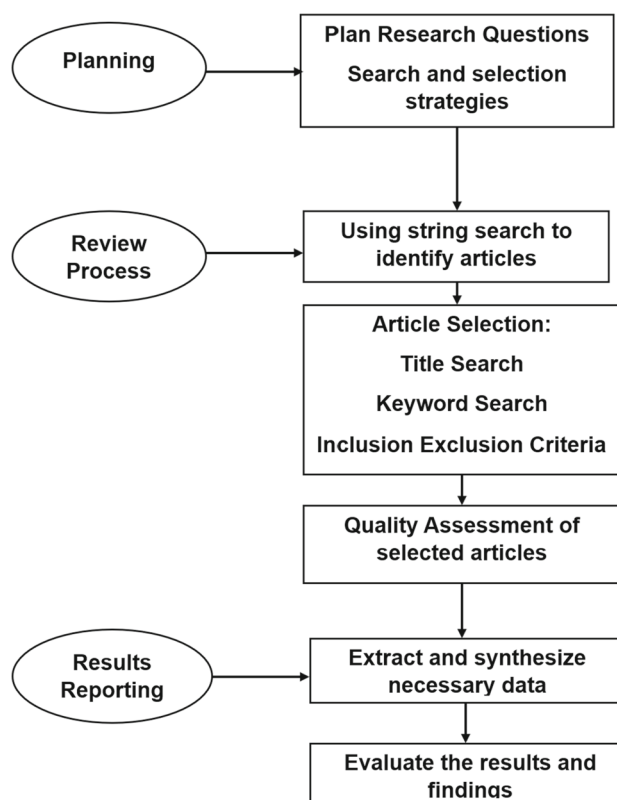


Fig. 1 An overview of the survey process

To effectively analyze the studies, first, the search and selection criteria are chosen, through which relevant articles are shortlisted. After that, the inclusion and exclusion criteria are decided; this is based on different ways such as title and string keyword search. Then, the quality assessment of each article is done, and a matrix is designed. Afterward, important information and results that align with the research questions are extracted and evaluated providing better insights.

The survey focuses on four main databases: IEEE, Springer, Science Direct, and PubMed, all of which are well-reputed and provide a good amount of articles. The articles chosen from these were those that meet the inclusion and exclusion criteria, and their quality is also evaluated. After comprehensively carrying out this process, the SLR concludes with effective and efficient studies that can assist in further research. The following review questions are extracted:

- *RQ-1*: What current challenges do radiologists face in generating accurate and correct medical reports?
- *RQ-2*: What are the current deep learning methods that enhance the generation of medical reports?
- *RQ-3*: How effective are deep learning-based methods in generating timely and accurate reports?
- *RQ-4*: What limitations do the current methods face and what are the future developments?

3.2 Reviewing process

After the process and strategies are planned in the planning phase, the review process is conducted; it contains the following steps:

3.3 Article search and selection criteria

The following databases are accessed to select articles for review.

- i. IEEE
- ii. Springer
- iii. ScienceDirect
- iv. PubMed

To collect relevant papers and articles from the databases, the string search method is used. The following collection of strings is searched:

- i. Medical report generation for image analysis.
- ii. Machine learning-based medical report generation.
- iii. Automated medical report generation using deep learning methods.
- iv. Image captioning and report generation.
- v. Medical report generation for chest x-rays.

- vi. Deep learning methods for CT image report generation.

This is an effective method to get a good collection of articles that can be further screened for relevant information. The main focus was on “medical report,” “deep learning,” and “image analysis.”

Various articles in the databases have been searched by different keywords relevant to medical report generation. These keywords are searched in the databases, and they follow the approach mentioned in [61] that uses Boolean terms such as AND and OR to combine different words and then search and select articles. So it is a combination of strings and boolean keywords. The following keywords are used for article search:

(“radiology” OR “report generation” OR “medical report generation” OR “radiologist”) AND (“natural language processing” OR “text mining” OR “information extraction” OR “document classification” OR “deep learning” OR “machine learning” OR “CNN” OR “RNN” OR “LSTM” OR “transformer” OR “attention” OR “GNN” OR “encoder-decoder” OR “LLM” OR “image analysis” OR “captioning”))

Title search

Titles relevant to “medical report generation,” “deep learning,” “CNN,” “RNN,” “LSTM,” “transformers,” “attention mechanism,” “GNN,” “LLMs,” “image prediction,” and “image analysis” are selected. Any article that does not contain these keywords is not selected.

3.4 Inclusion criteria

Different articles are selected based on the inclusion criteria [62]. First, a total of 1090 papers are searched from the four databases, and the filters are applied to them to select the most relevant papers.

Filters

To precisely choose the correct articles from the databases, different filters are applied. The following is the way the articles have been selected:

- Year: Articles published from the last 6 years from 2018 to 2024 are selected.
- Title: Only those articles containing titles relevant to medical report generation, deep learning methods, and weakly supervised object localization are chosen.
- Language: English-written articles have been evaluated.
- Quality: Articles that have good quality scores and that are relevant to the topic are selected.

After the filters are applied, only the important papers are left. Table 2 is a comprehensive analysis of how they are selected based on the database, title, language, year, and quality.

Table 2 Summary of papers

Category	Total	IEEE	Springer	Science Direct	PubMed
Total number of papers	1090	—	—	—	—
Papers in databases		520	290	180	100
Selection Criteria					
Title	550				
Language	200				
Year	120				
Quality	120				
Final	100				

3.5 Exclusion criteria

In the current survey, the articles are excluded based on the following criteria [63]:

- i. Articles that are not in the English language
- ii. Articles not relevant to medical report generation.
- iii. Redundant articles in databases.
- iv. All articles that are not from the years 2019–2024.
- v. Articles with a low impact factor are not selected.

3.6 Quality assessment

Quality is an important metric in determining the correctness of an article [63]. The survey assesses the quality of various articles based on the tool grading of recommendations, assessment, development, and evaluation (GRADE) mentioned in [64]. The GRADE methodology is a good and robust way of assessing the quality of research articles and provides a way for better recommendations in research to

enhance the report generation process. The main methodology of this approach is mentioned below:

- *Identify study designs*: This area focuses on the different types of studies that are selected such as how many articles are experiment-based and how many are review papers that focus on quantitative methods and qualitative methods.
- *Identify study designs*: The next step is assessing the bias of risk, which means to see if any of the studies are based or not on a particular approach and how that can impact the results.
- *Clarity and consistency*: How clear and concise the studies are as compared to the research questions, and how well the results are in each study conducted.
- *Directness*: How relevant the articles are to the goal of the research. How well it aligns with the specific data, reviews, and analysis.
- *Precision*: How reliable are the articles.
- *Publication bias*: This focuses on whether there is any bias in the publications of the studies that can hinder the process.

Table 3 shows an overview of the GRADE method applied to certain studies and its results.

Based on the results from this method, it is observed that the majority of the articles around 70% fall in the “high quality” category; this means that studies fall in this have minimum bias, correct methodology, clarity, and directness to the topic of research; the results are precise and are according to the research questions and nor bias in the publications as well. Additionally, around 20% articles fall under the “medium quality” category; they also provide profound results but have certain limitations and challenges. The remaining articles that are 10% fall under the “low quality” category, meaning they

Table 3 GRADE quality assessment for deep learning methods in medical report generation

Art.	Study design	Bias	Consistency	Directness	Precision	Pub. bias	Overall GRADE
[65]	RCT	Low	High	High	Mod	Low	High
[66]	Cohort	Low	Mod	High	High	Low	High
[67]	Case-control	High	Low	Mod	High	Low	Low
[68]	RCT	Mod	High	High	Low	High	Mod
[69]	Cohort	Mod	High	High	Mod	Low	High
[70]	Case-control	Mod	Low	Mod	High	Low	Low
[71]	Case-control	High	Low	Mod	High	Low	Low
[72]	Cohort	Low	Mod	High	High	Low	High
[73]	RCT	Low	High	High	High	Low	High
[74]	Cohort	Mod	High	High	Mod	Low	High
[75]	RCT	Mod	High	High	Low	High	Mod
[76]	Case-control	High	Low	Mod	High	Low	Low
[77]	RCT	Low	High	High	High	Low	High
[14]	Case-control	Mod	Low	Mod	High	Low	Low

Table 3 continued

Art.	Study design	Bias	Consistency	Directness	Precision	Pub. bias	Overall GRADE
[8]	RCT	Low	High	High	High	Low	High
[78]	Cohort	Mod	High	High	Low	High	Mod
[79]	Case–control	High	Low	Mod	High	Low	Low
[80]	Cohort	Low	High	High	Mod	Low	High
[81]	RCT	Low	High	High	High	Low	High
[82]	Case–control	High	Low	Mod	High	Low	Low
[83]	Cohort	Mod	High	High	Mod	Low	Mod
[84]	RCT	Mod	High	High	Low	High	Mod
[85]	Case–control	High	Low	Mod	High	Low	Low
[86]	Cohort	Low	Mod	High	High	Low	High
[87]	RCT	Low	High	High	High	Low	High
[88]	Case–control	Mod	High	High	High	Low	High

can be further enhanced to provide more robust and accurate results. This is demonstrated in Fig. 2.

Along with the GRADE approach, the articles are also evaluated based on some additional metrics. These provide valuable insights into the importance of each study and how impactful it is. The quality metrics that are chosen for this selection depend on the following:

- Database
- Impact factor
- Cite score
- Scopus score
- Web of Science score
- SJR score
- Quartile
- H-index

Each article is selected from a reliable source, removing the chance of incorrect or ambiguous ones. The results show that 90% of articles chosen have a good impact factor and cite score, meaning they have been used in existing research and are from good databases. Overall, the quality of the articles assessed is promising and can help in further research and

development. Table 4 presents the evaluation criteria for certain articles in this SLR; it is highlighted with certain points, while Table 5 shows the impact factor, h-index, quartile, etc. of a few selected articles. This is to make sure that good-quality articles are selected and reviewed.

4 Results and findings

This survey follows an empirical process that focuses on the collection of data and results and then evaluates them. This part of the paper focuses on the research questions that are extracted during the planning phase. These research questions are the basis of this study, as all the articles extracted along with them. This section focuses on the answers to the research questions while focusing on the studies. Table 6 contains the number of articles for each method of medical report generation.

4.1 RQ1-What current challenges do radiologists face in generating accurate and correct medical reports?

Before any automation, the traditional method to generate reports is through the expertise and medical knowledge of radiologists, who analyze images and then write detailed reports. The basic methods, as mentioned in [19], are intrinsic, extrinsic, text-to-text, human evaluation, and so on. However, they do face challenges why generating reports due to various reasons; the papers [98–102] discuss some of these issues. Table 7 discusses several challenges faced by radiologists.

The articles align correctly with this research question as they explore each and every aspect of why radiologists face challenges in generating accurate medical reports. These challenges then pave the way for advancements in medicine.

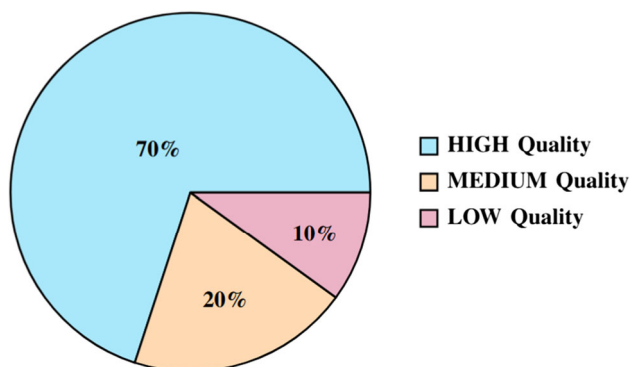
**Fig. 2** Distribution of article quality

Table 4 Quality assessment scores of various studies

Article ID	Clarity and coherence (20)	Depth of content (20)	Originality and creativity (15)	Accuracy and reliability (15)	Total score (100)
[1]	18	17	14	15	93
[6]	16	18	13	11	87
[10]	12	16	15	15	95
[33]	17	15	13	14	84
[75]	13	18	11	15	93
[13]	16	17	12	11	81
[69]	19	16	15	15	93
[89]	17	17	14	14	91
[78]	18	16	13	15	88
[45]	16	15	12	13	79
[66]	17	18	14	15	93
[32]	19	17	15	15	96
[87]	15	16	12	14	85
[19]	18	18	14	15	94
[41]	17	15	10	14	84

4.2 RQ2-What are the current deep learning methods that enhance the generation of medical reports?

Machine learning and deep learning methods provide solutions to the challenges that radiologists face. Since manually labeling and writing reports is time-consuming and expensive, it can also be expensive. Automated methods of report generation help solve this as reports can now be generated with little manual labeling effort. Numerous methods in deep learning such as encoder-decoder, transformers, RNN, LSTM, CNN, attention mechanism, LLMS, and graph-based models are introduced to generate timely and accurate medical reports for images.

The paper [27] proposes an encoder-decoder framework. The encoder extracts features and relevant information that

is a CNN, and the decoder and LSTM are responsible for generating textual reports. It adopts the process of transfer learning, and therefore it provides more flexibility and increased performance. Similar papers [89, 103–106] also focus on these models for encoder-decoder frameworks. Other methods than encoder-decoder are transformers that are widely used for medical report generation mentioned in papers [107–109]. These provide a robust way to generate clinically relevant and accurate reports.

By reviewing the articles, it is observed that around 45% use encoder-decoder frameworks, 20% transformer, and 10% for RNN-LSTM, attention mechanisms, and LLMS. Only 4 articles are found in the graph-based framework, and for non ml or traditional ml, one article is studied to gain further insights. Therefore, these results indicate that in medical report generation, the most commonly used method is the

Table 5 Various metrics of a few selected studies for analysis

Ref.	Journal	Scopus	Web of Science	Quartile	SJR	H-Index	Impact factor	Cite score
[8]	Bioengineering	Yes	Yes	Q2	0.633	46	4.6	4.2
[90]	Computerized Medical Imaging and Graphics	Yes	Yes	Q1	1.402	88	12.4	5.7
[91]	Journal of Biomedical Informatics	Yes	Yes	Q1	1.083	121	4.5	8.2
[92]	Medical Image Analysis	Yes	Yes	Q1	3.195	159	10.9	19.9
[93]	European Journal of Radiology	Yes	Yes	Q1	0.939	127	3.3	6.5
[94]	Computers in Biology and Medicine	Yes	Yes	Q1	1.22	113	7.7	9.2
[95]	Frontiers in Medicine	Yes	Yes	Q1	0.926	71	3.9	3.6
[96]	Physics in Medicine and Biology	Yes	Yes	Q1	0.865	206	3.5	6.9
[12]	Computers in Biology and Medicine	Yes	Yes	Q1	1.118	124	6.1	10.1
[97]	Yes	Yes	Yes	Q1	1.34	20	6.5	6.1

Table 6 Number of papers per technique

Techniques	Number of papers
Encoder-decoder	45
Transformers	20
RNN-LSTM	10
Attention mechanism	10
LLMs	10
Graph-based frameworks	4
Non-ML methods	1

encoder-decoder framework. Various levels of accuracy are reported in the existing works. Table 8 highlights the results of a few articles. From the data discussed in Table 8, it is evident that no one method is perfect and a combination of methods works best to improve the accuracy of report generation.

4.3 RQ3-What databases and datasets are accessed in the articles?

Table 9 displays the articles from various databases; after reviewing, the most accessed databases are IEEE, Springer, and PubMed, and the datasets NIH and cheXpert are widely used and contribute around 60% of the total datasets in the articles.

Table 10 displays the accuracies of certain studies highlighting that deep learning methods provide promising solutions in improving accuracy and timely report generation; however, these are not perfect results, and further research can improve their performance.

Figure 3 illustrates the comparison of deep learning models from the existing literature concerning accuracy, F1 score, etc. It indicates that several works report an accuracy higher than 90%. For example, [10, 11, 22] report a 92.1% accuracy. Similarly, accuracy achieved in [12, 41, 53, 54] is

Table 8 Number of papers per technique, accuracy scores, and outperformance

Techniques	Accuracy (%)	Outperformed
Encoder-decoder	86.2%	CNN-RNN [25]
Transformers	94.07%	HLSTM-VGG19 [6]
RNN-LSTM	93%	MFAT [42]
Attention mechanism	90.1%	[47]
LLMs	86.3%	GLM and ChatGLM [53]
Graph-based frameworks	82%	[55]

93.6%, 90.7%, 91.5%, and 92.1%, respectively. The study [23] reports the highest accuracy of 95.8%.

4.4 RQ4-What limitations do the current methods face and what are the future developments?

Automated report generation is a profound way of ensuring the correct and timely generation of reports; however, it faces certain limitations. The manual generation of reports faces challenges in accuracy, an important factor. The paper [111] mentions the challenges radiologists face when writing reports for medical images. It mentions various reasons that lead to such problems, and they are such as mismatch in patient results and miscommunication between physicians and radiologists. Due to this, reports are sometimes incorrectly generated, and then there is a need to correct and regenerate them, which is both costly and time-consuming.

Encoder-decoder frameworks [15, 112–114] are the most commonly used method in deep learning to generate medical reports, but they also face limitations due to training of data. These papers apply the model to single sources of data as well as multiple sources, and the limitations arise on the single resource data where training is done on one type of data that causes bias in the results and impacts the accuracy.

The studies [115–117] mention the use of transformers; they are often robust and more accurate as they capture com-

Table 7 Challenges faced by radiologists

Challenges	Details
1. Shortage of staff	• Fewer number of radiologists available leading to increased workload • Experienced staff is not available for serious cases
2. Maintenance of equipment	• Equipment malfunctioning leads to delays in results • Specialized equipment needs proper maintenance
3. Lack of communication	• Gap in communication between radiologists and physicians leads to unclear diagnosis and delayed treatment • Hindrance in obtaining proper clinical information
4. Inexperienced staff	• Staff with less experience leads to incorrect and delayed reports • Improper maintenance of EHR (Electronic Health Records)

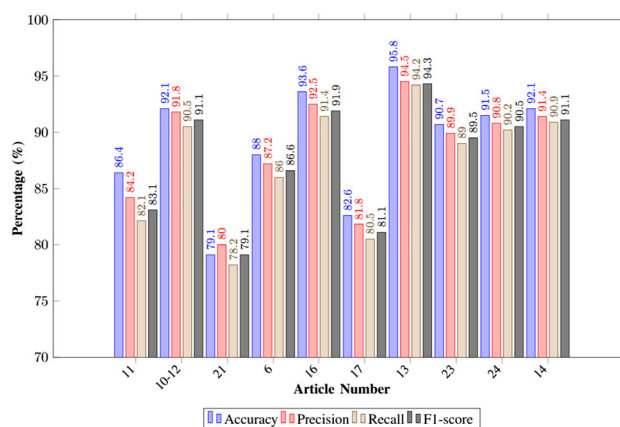
Table 9 Articles with corresponding databases and datasets

Reference	Databases	Datasets
[45]	IEEE, Springer	ChestX-ray14, IU-xray
[110]	Springer, Semantic Scholar	CheXpert
[83]	Semantic Scholar, Springer	MIMIC-CXR
[70]	IEEE, ScienceDirect	ChestX-ray8
[5]	ResearchGate, ACM Library	Open-I
[6]	Google Scholar, IEEE	NIH Chest X-rays
[105]	IEEE, ACM Library	COVID-19 Radiography
[8]	Springer, ScienceDirect	Shenzhen Hospital X-ray Set
[89]	Semantic Scholar	Montgomery County X-ray Set
[10]	ScienceDirect, Springer	VinDr-CXR, JSRT
[24]	ResearchGate	PadChest, Chexpert
[22]	Google Scholar	ChestX-ray14, MIMIC-CXR
[32]	IEEE	CheXpert, ChestX-ray8
[64]	Springer	MIMIC-CXR, VinDr-CXR
[40]	Semantic Scholar	ChestX-ray8, IU-xray

plex patterns in images but often struggle to capture high data dependencies and are computationally expensive. These limitations are the most found in the above papers.

After encoder-decoder frameworks, another similar approach is the use of RNN-based frameworks [118–120]. These models mostly focus on the textual report generation part, where a type of RNN, the LSTM, is used to generate meaningful sentences from words. So they are used in combination with neural networks for extraction of data from images such as CNN. The challenge these methods face is in the training; they work well on small datasets; however, as the size of the dataset increases, it becomes difficult for them to generate effective reports leading to overfitting.

A new deep learning method that is used to a great extent is attention mechanisms; these are combined with neural

**Fig. 3** Grouped bar chart for accuracy, precision, recall, and F1 score

network transformers and other deep learning models for generating medical reports. The papers [121, 122] mention a limitation faced by these methods, and that is they do not perform well on noisy data and can often hinder the accuracy.

In [53], the authors propose a solution for medical report generation using GPTS such as GLM and ChatGLM that use LLMS. However, these models face challenges when noise data is present and can lead to overfitting if the size of the dataset is large [54]. These limitations make the model more computationally expensive to implement.

Graph convolutional network also offers a promising approach to the generation of medical reports [55]. The limitation faced by these methods is that they are dependent on data quality; if good quality, it will lead to better results; however, if the quality is not up to the mark, it can hinder the accuracy of generating reports.

Based on the SLR conducted, this question is answered in Table 11. It illustrates the strengths and weaknesses of the methods that are studied in the literature.

The deep learning methods provide good and robust results, but they also face limitations and challenges that

Table 10 Summary of deep learning methods for medical report generation

Reference	DL model or technique	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
[11]	LSTM, GRU (Chexnet)	86.4	84.2	82.1	83.1
[10, 11, 22]	ConvNeXt (CNN), BioBERT	92.1	91.8	90.5	91.1
[24]	DenseNet-121, VLBERT	79.1	80.0	78.2	79.1
[6]	HLSTM, BERT embeddings	88.0	87.2	86.0	86.6
[41]	CNN (Inception-v3), LSTM	93.6	96.5	91.4	91.9
[42]	ResNet-152, transformer (MFAT)	82.6	81.8	80.5	81.1
[23]	AIMNet (transformer-based)	95.8	94.5	94.2	94.3
[53]	LoRA, P-tuning (LLM)	90.7	89.9	89.0	89.5
[54]	Swim transformer, LLM	91.5	90.8	90.2	90.5
[12]	Faster R-CNN, LSTM (SFNet)	92.1	91.4	90.9	91.1

Table 11 Overview of various technologies: pros and cons

Technology	Pros	Cons
Non-machine learning methods	Detailed reports with radiologist's expertise and medical knowledge	Time-consuming and expensive
Encoder-decoder frameworks	Reduces manual labeling of images and report generation	Biasness in high computational resource requirements
Transformer-based frameworks	Captures complex patterns and long-term dependencies	Computationally expensive and complex
RNN-LSTM frameworks	Handles sequential data for generating medical reports accurately	Difficult to train on large datasets can lead to overfitting
Attention-based frameworks	Captures both visual and semantic information from images, generating detailed reports	Depends on the quality of the dataset
Large language models	Generates well-structured and coherent reports, increases the performance of report generation	Faces challenges with noisy data
Graph-based frameworks	Uses graph-based methods such as GNNs to generate medically accurate reports	Depends on the quality of input data

leave a gap in the studies and can be further researched. These limitations can hinder the process of effectively generating medical reports. Future developments in this area are growing rapidly; studies [18, 90, 123–130] provide various methods to improve the process, one of which is to extract more discriminant features. Features are an important aspect in determining the complex patterns of images and provide better insights into the complexities. This can further enhance the research process.

Table 12 provides an analysis of medical report generation methods. The challenges mentioned in this study need

to be handled to ensure a more robust and precise report-generation process. Multi-model approaches can provide a good breakthrough as they incorporate information from various sources such as the combination of image data with lab notes, lab results, or the demographics and history of the patient. To further optimize, the process techniques such as LIME, SHAP, and LRP should be applied so the radiologists can trust AI-based decisions the models make to enhance the accuracy.

In real-world scenarios, radiologists often face difficulties in writing reports as it is time-consuming and costly.

Table 12 Comparative analysis of medical report generation methods

Method	Strengths	Weaknesses
Traditional methods	Reliable and based on radiologist expertise	Time-consuming and costly, with variability in results
Structural reporting	Ensures comprehensive coverage with standardized checklists	Limits radiologist individuality and flexibility
ChexNet (encoder-decoder)	High accuracy with reduced manual effort	Potential bias and high computational requirements
ConvNeXt + BioBERT	Strong context understanding in images and text	Limited long-range dependency capture and opaque decision-making
Medical-VLBERT	Effective for diagnosing diseases with transfer learning	Extensive data preprocessing needed; struggles with image intricacies
DenseNet121 + GRU	Good feature extraction from images for report generation	Challenges with complex medical terms and computational intensity
AIMNet	Tailored reports using advanced attention mechanisms	Complexity and potential bias from training data
Transformer-based models	High accuracy and dependency capture	Computationally expensive and risk of overfitting
SFNet (semantic fusion network)	Combines lesion detection with report generation for better accuracy	Potential inaccuracy in lesion detection due to training data bias
Encoder-decoder with CNNs	Streamlines workflow and reduces radiologist burden	Long reports and bias in embeddings can affect report quality
ResNet-152 with transformer decoder	Improves text flow and handles class imbalance effectively	Increased training time due to MFAT may require optimization

Table 12 continued

Method	Strengths	Weaknesses
Flexible multi-view paradigm (FMVP)	Enhances accuracy and eases radiologist burden through retrieval methods	Requires further clinical validation despite good NLG metrics
RNN-LSTM frameworks	Effective at capturing sequential patterns for detailed reports	Struggles with large datasets leading to overfitting
Attention-based frameworks	Captures visual and semantic features effectively	Complexity and reliance on high-quality annotations
Semantic inspector	Identifies important regions in images for better reports	High dependence on dataset quality
Hierarchical attention system	Enhances understanding of text with spatial and channel information	Increased training time due to complexity
SKEMA framework	Integrates visual and semantic information	Quality of knowledge graph affects accuracy
Attention guided object dropout MLP	Focuses on relevant features for diagnosis	Challenges in generalizability and resource intensity

The addition of automated methods for report generation has paved a more robust and precise way. However, integrating it into real-world medical workflows has its strengths and weaknesses. The developed technology needs to be easy to use for the radiologists to adapt to it without requiring rigorous training. Additionally, to improve patient care, the report turnaround times must be efficient and precise.

Medical information is highly sensitive; therefore, patient data is confidential. When developing AI-based systems, it is crucial to keep in my ethical considerations for applications in healthcare. Developer must make sure that their systems have strong data security features that are in conformity with international standards. HIPPA (Health Insurance Portability and Accountability Act) is one such healthcare legislation that is responsible for maintaining the security of patient information.

4.5 Imbalanced datasets

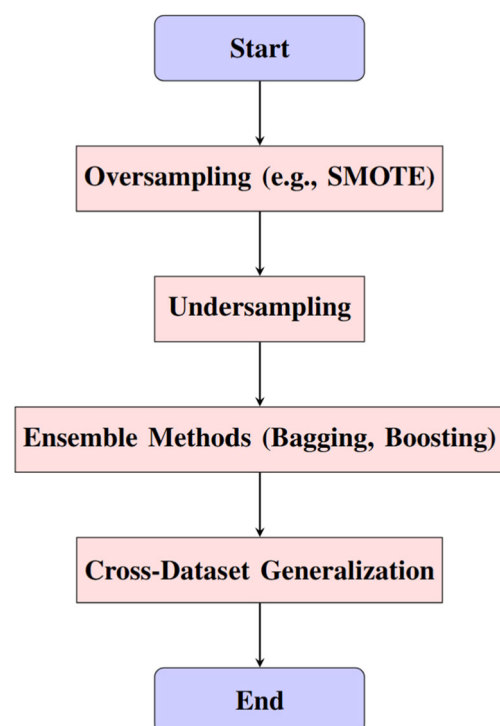
An imbalanced dataset occurs when classes are unevenly distributed, leading to a situation where one class outnumbers another class. This can greatly impact the performance of the model in classification tasks and can lead to model biases and overfitting. Addressing this is a crucial step in classification tasks to ensure that the models are accurate and precise. The following figure showcases certain methods that are widely used to handle imbalances in datasets:

- *Oversampling*: Techniques such as SMOTE.
- *Undersampling*: Strategies for reducing majority class samples.
- *Ensemble methods*: Use of bagging and boosting to improve performance.
- *Cross-dataset generalization*: Approaches to enhance robustness when dealing with varying dataset characteristics.

Figure 4 visually summarizes the methods, showing how each technique can be applied to mitigate the effects of imbalanced datasets.

4.6 Explainable AI

AI-driven decisions need to be clear and well understood by clinical physicians to make sure medical applications and diagnoses aid in the better development of patients. Explainable AI (XAI) [59] ensures that the reason behind such

**Fig. 4** Data resampling approaches for imbalanced datasets

decisions is precisely interpreted and analyzed. XAI techniques such as LRP, SHAP, and LIME [128] help to explain the predictions made by models made by AI. It is an important part of crucial fields, especially in medical image analysis with a focus on optimizing and improving diagnosis and report generation. Figure 5 shows a comparative analysis of various XAI techniques.

4.7 Sensitive parameters affecting image processing diagnosis

Several parameters affect the diagnosis of a disease, so understanding them is crucial for optimal and accurate patient results. The following are certain parameters:

4.7.1 Image acquisition parameters

- **Resolution:** When the resolution of images is higher, it enhances the detail and accuracy, but the file size and processing time increases.
- **Noise:** It is important to reduce noise in an image to enhance clarity and abstain from hiding crucial details.

4.7.2 Image preprocessing techniques

- **Segmentation:** Proper image segmentation reduces diagnostic errors
- **Normalization:** Alters image intensity providing consistency.
- **Feature extraction:** The correct feature extraction enhances algorithm performance and model accuracy.

4.7.3 Image analysis considerations

- **Parameter tuning:** The right parameters optimize model performance.
- **Classifier selection:** Classifiers impact the accuracy and performance.

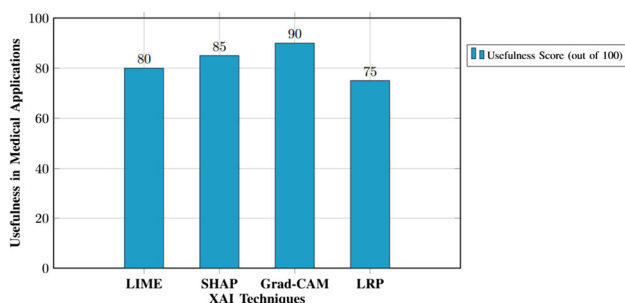


Fig. 5 Comparison of different XAI techniques in medical applications

4.7.4 Disease-specific factors

- **Image modality:** Various algorithms impact different image modalities.
- **Clinical context:** For more accurate reports, specific patient information is crucial.

An accuracy of 98% is ideal for the correct diagnosis of any disease; however, the literature studied for this study comes close to this threshold, but it still is a challenge to achieve this goal. This concludes that there is still further room for improvement and research so an optimal method can be developed to maintain this accuracy.

The tools for developing and implementing the algorithms are as follows:

- Python, tensor flow, PyTorch, Keras
- Natural Language Processing
- Transformers (BERT, GPT)
- Attention Mechanism
- CNNs
- RNNs
- GNNs
- Multi-model methods
- LRP, SHAP, LIME
- ResNet, VGG,
- LLMS
- LSTM

The tools for drawing the diagrams:

- Lucid Chart
- Latex for diagrams

5 Conclusion and future work

The inclusion of deep learning methods has certainly improved the process of medical report generation by automating the labeling process of images and the writing of textual reports. This study conducts a survey on various methods and how they impact the process of medical report generation. The application of these methods streamlines the workflow and reduces the burden on radiologists. They also help in standardizing medical reports by improving clarity and accuracy. Due to these applications, it helps in faster and improved patient outcomes, bridging the gap between people and healthcare, especially in areas where resources are limited, and reducing the cost and labor due to extensive manual labeling and writing. On the basis of the studies, the most commonly applied methods are encoder-decoder frameworks and transformers, followed by recurrent

neural networks which are the second most applied techniques. In addition, attention mechanisms, large language models, and graph-based methods are also among the widely adopted approaches. Despite advancements in medical image analysis, including image captioning and report generation, existing models suffer from poor performance when dealing with images containing complex medical details. Similarly, deep learning models experience bias due to imbalanced and poor-quality datasets. Such problems persist often when more discriminate features are needed from complex medical images. However, even though automated medical report generation provides a breakthrough in the field of medicine, it is important to address ethical concerns such as privacy of patient data, biases in data, and accountability for the decisions made. This study also pointed out certain limitations of these methods, and further research and development are required to improve their performance. For future work in this domain, different methods of extracting discriminant features can be applied to gather more valuable patterns of images that are complex to understand. Furthermore, advanced deep learning models capable of obtaining better and more complex details from medical images are needed to enhance the accuracy of the report generation process. Combining different deep learning models to improve accuracy and implementing these methods in real-world applications can also be a potential avenue for further research.

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Availability of data and materials There is no data associated with this study.

Declarations

Conflict of interest The authors declare no competing interests.

References

- Li R, Wang Z, Zhang L (2021) Image caption and medical report generation based on deep learning: a review and algorithm analysis. In: 2021 International Conference on Computer Information Science and Artificial Intelligence (CISAI), pp 373–379. IEEE
- Rinaldi AM, Russo C, Tommasino C (2023) Automatic image captioning combining natural language processing and deep neural networks. *Results Eng* 18:101107
- Al-Malla MA, Jafar A, Ghneim N (2022) Image captioning model using attention and object features to mimic human image understanding. *J Big Data* 9(1):20
- Gupta SC, Singh NR, Sharma T, Tyagi A, Majumdar R (2021) Generating image captions using deep learning and natural language processing. In: 2021 9th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions)(ICRITO), pp 1–4. IEEE
- Sehgal S, Sharma J, Chaudhary N (2020) Generating image captions based on deep learning and natural language processing. In: 2020 8th International Conference on Reliability, Infocom Technologies and Optimization (Trends and Future Directions)(ICRITO), pp 165–169. IEEE
- Rohitharun S, Reddy LUK, Sujana S (2022) Image captioning using CNN and RNN. In: 2022 2nd Asian Conference on Innovation in Technology (ASIANCON), pp 1–8. IEEE
- Alfarghaly O, Khaled R, Elkorany A, Helal M, Fahmy A (2021) Automated radiology report generation using conditioned transformers. *Inf Med Unlocked* 24:100557
- Gu Y, Li R, Wang X, Zhou Z (2023) Automatic medical report generation based on cross-view attention and visual-semantic long short term memory. *Bioengineering* 10(8):966
- Yin C, Qian B, Wei J, Li X, Zhang X, Li Y, Zheng Q (2019) Automatic generation of medical imaging diagnostic report with hierarchical recurrent neural network. In: 2019 IEEE International Conference on Data Mining (ICDM), pp 728–737. IEEE
- Sirshar M, Paracha MFK, Akram MU, Alghamdi NS, Zaidi SZY, Fatima T (2022) Attention based automated radiology report generation using CNN and LSTM. *PLoS One* 17(1):0262209
- Kumar MA, Panitini M, Vemulapalli S, Sai MJNV (2023) Deep learning based automatic radiology report generation. In: 2023 Third International Conference on Artificial Intelligence and Smart Energy (ICAIS), pp 1521–1526. IEEE
- Zeng X, Wen L, Xu Y, Ji C (2020) Generating diagnostic report for medical image by high-middle-level visual information incorporation on double deep learning models. *Comput Methods Prog Biomed* 197:105700
- Liu Z, Zhu Z, Zheng S, Zhao Y, He K, Zhao Y (2023) From observation to concept: a flexible multi-view paradigm for medical report generation. *IEEE Trans Multimed*
- Pang T, Li P, Zhao L (2023) A survey on automatic generation of medical imaging reports based on deep learning. *BioMed Eng OnLine* 22(1):48
- Monshi MMA, Poon J, Chung V (2020) Deep learning in generating radiology reports: a survey. *Artif Intell Med* 106:101878
- Beddiar D-R, Ouassalah M, Seppänen T (2023) Automatic captioning for medical imaging (MIC): a rapid review of literature. *Artif Intell Rev* 56(5):4019–4076
- Wang Z, Han H, Wang L, Li X, Zhou L (2022) Automated radiographic report generation purely on transformer: a multicriteria supervised approach. *IEEE Trans Med Imaging* 41(10):2803–2813
- Xi E (2021) Image feature extraction and analysis algorithm based on multi-level neural network. In: 2021 5th International Conference on Computing Methodologies and Communication (ICCMC), pp 1062–1065. IEEE
- Zhou Y, Ringeval F, Portet F (2023) A survey of evaluation methods of generated medical textual reports. In: Proceedings of the 5th clinical natural language processing workshop, pp 447–459
- Ramsay CR, Grant AM, Wallace SA, Garthwaite PH, Monk AF, Russell IT (2000) Assessment of the learning curve in health technologies: a systematic review. *Int J Technol Assess Health Care* 16(04):1095–1108
- Granata V, De Muzio F, Cutolo C, Dell'Aversana F, Grassi F, Grassi R, Simonetti I, Bruno F, Palumbo P, Chiti G et al (2022)

- Structured reporting in radiological settings: pitfalls and perspectives. *J Personal Med* 12(8):1344
22. Akbar W, Haq MIU, Soomro A, Daudpota SM, Imran AS, Ullah M (2023) Automated report generation: a GRU based method for chest X-rays. In: 2023 4th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET), pp 1–6. IEEE
 23. Shi J, Wang S, Wang R, Ma S (2022) Aimnet: adaptive image-tag merging network for automatic medical report generation. In: ICASSP 2022–2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 7737–7741. IEEE
 24. Kitchenham B, Brereton OP, Budgen D, Turner M, Bailey J, Linkman S (2009) Systematic literature reviews in software engineering—a systematic literature review. *Inf Softw Technol* 51(1):7–15
 25. Babar Z, Laarhoven T, Marchiori E (2021) Encoder-decoder models for chest X-ray report generation perform no better than unconditioned baselines. *PLoS One* 16(11):0259639
 26. Yang Y, Yu J, Zhang J, Han W, Jiang H, Huang Q (2021) Joint embedding of deep visual and semantic features for medical image report generation. *IEEE Trans Multimed* 25:167–178
 27. Li M, Liu R, Wang F, Chang X, Liang X (2023) Auxiliary signal-guided knowledge encoder-decoder for medical report generation. *World Wide Web* 26(1):253–270
 28. Zhang S, Han Q, Li J, Sun Y, Qin Y (2024) A medical report generation method integrating teacher-student model and encoder-decoder network. *Biomed Signal Process Control* 94:106251
 29. Yang S, Ji J, Zhang X, Liu Y, Wang Z (2021) Weakly guided hierarchical encoder-decoder network for brain CT report generation. In: 2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp 568–573. IEEE
 30. Pandey S, Saha P, Sharan G, Sandosh S (2024) Enhancing chest X-ray analysis using encoder-decoder with GRU for report generation. In: 2024 Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT), pp 1–8. IEEE
 31. Amjoud AB, Amrouch M (2021) Automatic generation of chest X-ray reports using a transformer-based deep learning model. In: 2021 Fifth International Conference On Intelligent Computing in Data Sciences (ICDS), pp 1–5. IEEE
 32. Huang Z, Zhang X, Zhang S (2023) Kiut: knowledge-injected u-transformer for radiology report generation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 19809–19818
 33. Tsaniya H, Fatichah C, Suciati N (2024) Automatic radiology report generator using transformer with contrast-based image enhancement. *IEEE Access*
 34. Mohsan MM, Akram MU, Rasool G, Alghamdi NS, Baqai MAA, Abbas M (2022) Vision transformer and language model based radiology report generation. *IEEE Access* 11:1814–1824
 35. Xu L, Tang Q, Zheng B, Lv J, Li W, Zeng X (2024) CGFTrans: cross-modal global feature fusion transformer for medical report generation. *IEEE J Biomed Health Inf*
 36. Kim J, Kim BS, Choi I, Yang Z, Jang B (2024) FTT: Fourier transform based transformer for brain CT report generation. In: 2024 International Conference on Artificial Intelligence in Information and Communication (ICAICI), pp 617–621. IEEE
 37. Mondal C, Pham D-S, Tan T, Gedeon T, Gupta A (2023) Transformers are all you need to generate automatic report from chest X-ray images. In: 2023 International Conference on Digital Image Computing: Techniques and Applications (DICTA), pp 387–394. IEEE
 38. Yi X, Fu Y, Liu R, Zhang H, Hua R (2024) TSGET: two-stage global enhanced transformer for automatic radiology report generation. *IEEE J Biomed Health Inf*
 39. Nguyen HT, Nie D, Badamdorj T, Liu Y, Hong L, Truong J, Cheng L (2022) Eddie-transformer: enriched disease embedding transformer for X-ray report generation. In: 2022 IEEE 19th International Symposium on Biomedical Imaging (ISBI), pp 1–5. IEEE
 40. Kumar MA, Ganta S, Chinni GR (2023) Report generation on chest X-rays using deep learning. In: 2023 7th International Conference on Intelligent Computing and Control Systems (ICICCS), pp 376–381. IEEE
 41. Singh S, Karimi S, Ho-Shon K, Hamey L (2019) From chest X-rays to radiology reports: a multimodal machine learning approach. In: 2019 Digital Image Computing: Techniques and Applications (DICTA), pp 1–8. IEEE
 42. Yu H, Zhang Q (2022) Clinically coherent radiology report generation with imbalanced chest X-rays. In: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp 1781–1786. IEEE
 43. Ram KB, Venkatesh B, Sree SPS, Anilkumar C, Reddy VSN, Kodumuri B (2023) Image caption and speech generation using LSTM and GTTS API. In: 2023 Second International Conference on Augmented Intelligence and Sustainable Systems (ICAISS), pp 992–997. IEEE
 44. Shehzad MK, Rose L, Assaad M (2021) Rnn-based twin channel predictors for CSI acquisition in UAV-assisted 5G+ networks. In: 2021 IEEE Global Communications Conference (GLOBECOM), pp 1–6. IEEE
 45. Sreedevi B, Amirthavarshini T, Anitha S, Shwetha G (2022) Web based disease prediction and forecasting with KNN and RNN using internet of medical things. In: 2022 International Conference on Computer, Power and Communications (ICCP), pp 192–198. IEEE
 46. Reddy PM, Verma VK, Varma MVC (2023) Optimizing medical image report generation with varied attention mechanisms. In: 2023 6th International Conference on Contemporary Computing and Informatics (IC3I), vol 6, pp 2137–2143. IEEE
 47. Kisilev P, Walach E, Barkan E, Ophir B, Alpert S, Hashoul SY (2015) From medical image to automatic medical report generation. *IBM J Res Dev* 59(2/3):2–1
 48. Lin Y, Tang Q, Wang H, Huang C, Favour E, Wang X, Feng X, Yu Y (2023) Attention enhanced network with semantic inspector for medical image report generation. In: 2023 IEEE 35th International Conference on Tools with Artificial Intelligence (ICTAI), pp 242–249. IEEE
 49. Huang X, Yan F, Xu W, Li M (2019) Multi-attention and incorporating background information model for chest X-ray image report generation. *IEEE Access* 7:154808–154817
 50. Xu D, Chen Y, Zhang J, Lou Y, Wang H, He J, Huang Y (2023) Radiology report generation via structured knowledge-enhanced multi-modal attention and contrastive learning. In: 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp 2320–2325. IEEE
 51. Chen Z, Tang Y (2022) Improving radiology report generation via object dropout strategy and MLP-based captioner. In: 2022 IEEE 5th Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC), vol 5, pp 316–322. IEEE
 52. Itoh TD, Kubo T, Ikeda K, Maruno Y, Ikutani Y, Hata H, Matsumoto K, Ikeda K (2019) Towards generation of visual attention map for source code. In: 2019 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), pp 951–954. IEEE
 53. Wu W, Li M, Wu J, Ni M, Yuan H (2023) Learning to generate radiology findings from impressions based on large language model. In: 2023 IEEE international conference on Big Data (BigData), pp 2550–2554. IEEE
 54. Wang Z, Liu L, Wang L, Zhou L (2023) R2GengPT: radiology report generation with frozen LLMs. *Meta-Radiol* 1(3):100033

55. Zhang D, Ren A, Liang J, Liu Q, Wang H, Ma Y (2022) Improving medical X-ray report generation by using knowledge graph. *Appl Sci* 12(21):11111
56. Moon JH, Lee H, Shin W, Kim Y-H, Choi E (2022) Multi-modal understanding and generation for medical images and text via vision-language pre-training. *IEEE J Biomed Health Inf* 26(12):6070–6080
57. Zhang K, Yang Y, Yu J, Fan J, Jiang H, Huang Q, Han W (2024) Attribute prototype-guided iterative scene graph for explainable radiology report generation. *IEEE Trans Med Imaging*
58. Chen J, Pan R (2024) Medical report generation based on multimodal federated learning. *Comput Med Imaging Graph* 113:102342
59. Ahmed SB, Solis-Oba R, Ilie L (2022) Explainable-AI in automated medical report generation using chest X-ray images. *Appl Sci* 12(22):11750
60. Band SS, Yarahmadi A, Hsu C-C, Biyari M, Sookhak M, Ameri R, Dehzangi I, Chronopoulos AT, Liang H-W (2023) Application of explainable artificial intelligence in medical health: a systematic review of interpretability methods. *Inf Med Unlocked* 40:101286
61. Torres-Carrión PV, González-González CS, Aciar S, Rodríguez-Morales G (2018) Methodology for systematic literature review applied to engineering and education. In: 2018 IEEE Global Engineering Education Conference (EDUCON), pp 1364–1373. IEEE
62. Chang C-Y, Lai C-L, Hwang G-J (2018) Trends and research issues of mobile learning studies in nursing education: a review of academic publications from 1971 to 2016. *Comput Educ* 116:28–48
63. Bezerra CT, Grande AJ, Galvão VK, Santos DHMD, Atallah ÁN, Silva V (2022) Assessment of the strength of recommendation and quality of evidence: grade checklist. A descriptive study. *Sao Paulo Med J* 140(6):829–836
64. Ennadifi E, Laraba S, Vincke D, Mercatoris B, Gosselin B (2020) Wheat diseases classification and localization using convolutional neural networks and GradCAM visualization. In: 2020 International Conference on Intelligent Systems and Computer Vision (ISCV), pp 1–5. IEEE
65. Zhu Y, Zhou Y, Ye Q, Qiu Q, Jiao J (2017) Soft proposal networks for weakly supervised object localization. In: Proceedings of the IEEE international conference on computer vision, pp 1841–1850
66. Chen L, Zhang H, Xiao J, Nie L, Shao J, Liu W, Chua T-S (2017) SCA-CNN: spatial and channel-wise attention in convolutional networks for image captioning. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 5659–5667
67. Inbaraj XA, Villavicencio C, Macrohon JJ, Jeng J-H, Hsieh J-G (2021) Object identification and localization using Grad-CAM++ with mask regional convolution neural network. *Electronics* 10(13):1541
68. Ma Y, Ma AJ, Pan Y, Chen X (2020) Multi-scale feature pyramids for weakly supervised thoracic disease localization. In: 2020 IEEE International Conference on Image Processing (ICIP), pp 2481–2485. IEEE
69. Yang Z, Zhao L, Wu S, Chen CY-C (2021) Lung lesion localization of COVID-19 from chest CT image: a novel weakly supervised learning method. *IEEE J Biomed Health Inf* 25(6):1864–1872
70. Ouyang X, Karanam S, Wu Z, Chen T, Huo J, Zhou XS, Wang Q, Cheng J-Z (2020) Learning hierarchical attention for weakly-supervised chest X-ray abnormality localization and diagnosis. *IEEE Trans Med Imaging* 40(10):2698–2710
71. Lu J, Xiong C, Parikh D, Socher R (2017) Knowing when to look: adaptive attention via a visual sentinel for image captioning. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 375–383
72. Soofi AA et al (2023) Exploring deep learning techniques for glaucoma detection: a comprehensive review. [arXiv:2311.01425](https://arxiv.org/abs/2311.01425)
73. Zhang Z, Chang M-C, Bui TD (2022) Improving class activation map for weakly supervised object localization. In: ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp 2624–2628. IEEE
74. Gulum MA, Trombley CM, Kantardzic M (2021) Improved deep learning explanations for prostate lesion classification through Grad-CAM and saliency map fusion. In: 2021 IEEE 34th International Symposium on Computer-Based Medical Systems (CBMS), pp 498–502. IEEE
75. Mao C, Yao L, Luo Y (2022) ImageGCN: multi-relational image graph convolutional networks for disease identification with chest X-rays. *IEEE Trans Med Imaging* 41(8):1990–2003
76. Yuan Z, Li X, Wang Q (2019) Exploring multi-level attention and semantic relationship for remote sensing image captioning. *IEEE Access* 8:2608–2620
77. Davidson EM, Poon MT, Casey A, Grivas A, Duma D, Dong H, Suárez-Paniagua V, Grover C, Tobin R, Whalley H et al (2021) The reporting quality of natural language processing studies: systematic review of studies of radiology reports. *BMC Med Imaging* 21:1–13
78. Alqahtani FF, Mohsan MM, Alshamrani K, Zeb J, Alhamami S, Alqarni D (2024) CNX-B2: a novel CNN-transformer approach for chest X-ray medical report generation. *IEEE Access* 12:26626–26635
79. Xia R, Li G, Huang Z, Wen L, Pang Y (2021) Classify and localize threat items in X-ray imagery with multiple attention mechanism and high-resolution and high-semantic features. *IEEE Trans Instrum Meas* 70:1–10
80. Sriker D, Greenspan H, Goldberger J (2022) Class-based attention mechanism for chest radiograph multi-label categorization. In: 2022 IEEE 19th International Symposium on Biomedical Imaging (ISBI), pp 1–5. IEEE
81. Zhao G, Zhao Z, Gong W, Li F (2023) Radiology report generation with medical knowledge and multilevel image-report alignment: a new method and its verification. *Artif Intell Med* 146:102714
82. Xiao M, Zhang L, Shi W, Liu J, He W, Jiang Z (2021) A visualization method based on the Grad-CAM for medical image segmentation model. In: 2021 International Conference on Electronic Information Engineering and Computer Science (EIECS), pp 242–247. IEEE
83. Zhang S, Tan L, Han Q, Wang H, Meng J (2023) Automatic report generation on a large-scale stroke MRI dataset. In: 2023 IEEE 6th International Conference on Electronic Information and Communication Technology (ICEICT), pp 123–128. IEEE
84. Vimala R, Preethi D (2024) Maxillary sinus disease detection and analysis approaches in deep learning: survey. In: 2024 International Conference on Cognitive Robotics and Intelligent Systems (ICC-ROBINS), pp 173–181. IEEE
85. Wu J, Gur Y, Karargyris A, Syed AB, Boyko O, Moradi M, Syeda-Mahmood T (2020) Automatic bounding box annotation of chest X-ray data for localization of abnormalities. In: 2020 IEEE 17th International Symposium on Biomedical Imaging (ISBI), pp 799–803. IEEE
86. Lan S, Zhou C, Chen L, Fan H, Yan N, Huang Y (2021) Automatic report generation based on multi-modal and multi-view model for fundus images. In: 2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp 945–951. IEEE
87. Hou D, Zhao Z, Liu Y, Chang F, Hu S (2021) Automatic report generation for chest X-ray images via adversarial reinforcement learning. *IEEE Access* 9:21236–21250
88. Chien J-C, Lee J-D, Hu C-S, Wu C-T (2022) The usefulness of gradient-weighted CAM in assisting medical diagnoses. *Appl Sci* 12(15):7748
89. Nguyen E, Theodorakopoulos D, Pathak S, Geerdink J, Vijlbrieff O, Van Keulen M, Seifert C (2020) A hybrid text classification and language generation model for automated summarization

- of Dutch breast cancer radiology reports. In: 2020 IEEE Second International Conference on Cognitive Machine Intelligence (CogMI), pp 72–81. IEEE
90. Ouis MY, Akhloufi M (2023) Deep learning for report generation on chest X-ray images. *Comput Med Imaging Graph* 102320
 91. Kaur N, Mittal A (2022) RadioBERT: a deep learning-based system for medical report generation from chest X-ray images using contextual embeddings. *J Biomed Inf* 135:104220
 92. Shen Y, Wu N, Phang J, Park J, Liu K, Tyagi S, Heacock L, Kim SG, Moy L, Cho K et al (2021) An interpretable classifier for high-resolution breast cancer screening images utilizing weakly supervised localization. *Med Image Anal* 68:101908
 93. Neves J, Hsieh C, Nobre IB, Sousa SC, Ouyang C, Maciel A, Duchowski A, Jorge J, Moreira C (2024) Shedding light on ai in radiology: a systematic review and taxonomy of eye gaze-driven interpretability in deep learning. *Eur J Radiol* 111341
 94. Kuang Z, Yan Z, Yu L (2024) Weakly supervised learning for multi-class medical image segmentation via feature decomposition. *Comput Biol Med* 171:108228
 95. Albahli S, Nazir T (2022) AI-CenterNet CXR: an artificial intelligence (AI) enabled system for localization and classification of chest X-ray disease. *Front Med* 9:955765
 96. Wang J, Zhao H, Liang W, Wang S, Zhang Y (2023) Cross-convolutional transformer for automated multi-organs segmentation in a variety of medical images. *Phys Med Biol* 68(3):035008
 97. Yu F, Endo M, Krishnan R, Pan I, Tsai A, Reis EP, Fonseca EKUN, Lee HMH, Abad ZSH, Ng AY et al (2023) Evaluating progress in automatic chest X-ray radiology report generation. *Patterns* 4(9)
 98. Hinrichs-Krapels S, Tombo L, Boulding H, Majonga ED, Cummins C, Manaseki-Holland S (2023) Barriers and facilitators for the provision of radiology services in Zimbabwe
 99. Pesapane F, Tantrige P, De Marco P, Carriero S, Zugni F, Nicosia L, Bozzini AC, Rotili A, Latronico A, Abbate F et al (2023) Advancements in standardizing radiological reports: a comprehensive review. *Medicina* 59(9):1679
 100. Woznitza N, Steele R, Groombridge H, Compton E, Gower S, Hussain A, Norman H, O'Brien A, Robertson K (2021) Clinical reporting of radiographs by radiographers: policy and practice guidance for regional imaging networks. *Radiography* 27(2):645–649
 101. Thompson JD (2022) Toward consistent design and reporting of observer studies in imaging. *Radiological Society of North America*
 102. Lundvall L-LC, Dahlström N, Dahlgren MA (2021) Professional challenges in medical imaging for providing safe medical service. *Prof Professionalism* 11(2)
 103. Zeng L, Zhang X, Wang T, Li X, Yu J, Wang H (2018) Improving code summarization by combining deep learning and empirical knowledge (s). In: SEKE, pp 566–565
 104. Tagawa Y, Shimada K (2017) Simple and sophisticated innig summary generation based on encoder-decoder model and transfer learning. In: 2017 International Conference on Asian Language Processing (IALP), pp 252–255. IEEE
 105. Agrawal T, Choudhary P (2023) COVID-SegNet: encoder-decoder-based architecture for COVID-19 lesion segmentation in chest X-ray. *Multimed Syst* 29(4):2111–2124
 106. Sreelakshmi S, Malu G, Sherly E, Mathew R (2023) M-Net: an encoder-decoder architecture for medical image analysis using ensemble learning. *Results Eng* 17:100927
 107. Wang H, Niu J, Liu X, Wang Y (2022) Embracing uniqueness: generating radiology reports via a transformer with graph-based distinctive attention. In: 2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM), pp 581–588. IEEE
 108. Wang Z, Liu L, Wang L, Zhou L (2023) METransformer: radiology report generation by transformer with multiple learnable expert tokens. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 11558–11567
 109. Nerella S, Bandyopadhyay S, Zhang J, Contreras M, Siegel S, Bumin A, Silva B, Sena J, Shickel B, Bihorac A et al (2023) Transformers in healthcare: a survey. [arXiv:2307.00067](https://arxiv.org/abs/2307.00067)
 110. Van Hai V, Le Thi Kim Nhung H, Hoc HT (2019) A review of software effort estimation by using functional points analysis. *Computational statistics and mathematical modeling methods in intelligent systems: Proceedings of 3rd computational methods in systems and software 2019*, vol 2 3, pp 408–422
 111. Sohoni CA (2013) Medical negligence: a difficult challenge for radiology. *Ind J Radiol Imaging* 23(01):110–112
 112. Banerjee I, Ling Y, Chen MC, Hasan SA, Langlotz CP, Moradzadeh N, Chapman B, Amrhein T, Mong D, Rubin DL et al (2019) Comparative effectiveness of convolutional neural network (CNN) and recurrent neural network (RNN) architectures for radiology text report classification. *Artif Intell Med* 97:79–88
 113. Kanjanasurat I, Tenghongsakul K, Purahong B, Lasakul A (2023) CNN-RNN network integration for the diagnosis of COVID-19 using chest X-ray and CT images. *Sensors* 23(3):1356
 114. Yu J, Antonio A, Villalba-Mora E (2022) Deep learning (CNN, RNN) applications for smart homes: a systematic review. *Computers* 11(2):26
 115. Kong M, Huang Z, Kuang K, Zhu Q, Wu F (2022) Transq: transformer-based semantic query for medical report generation. In: International conference on medical image computing and computer-assisted intervention, pp 610–620. Springer
 116. Cao Y, Cui L, Zhang L, Yu F, Li Z, Xu Y (2023) MMTN: multi-modal memory transformer network for image-report consistent medical report generation. *Proceedings of the AAAI conference on artificial intelligence* 37:277–285
 117. Cao Y, Cui L, Yu F, Zhang L, Li Z, Liu N, Xu Y (2022) KdTNet: medical image report generation via knowledge-driven transformer. In: International conference on database systems for advanced applications, pp 117–132. Springer
 118. Ordu SK, Yıldız O (2023) Summarizing medical imaging reports with LSTM deep learning method for effective diagnosis/treatment process
 119. Yang X, Ye M, You Q, Ma F (2021) Writing by memorizing: hierarchical retrieval-based medical report generation. [arXiv:2106.06471](https://arxiv.org/abs/2106.06471)
 120. Liu F, Ge S, Zou Y, Wu X (2022) Competence-based multimodal curriculum learning for medical report generation. [arXiv:2206.14579](https://arxiv.org/abs/2206.14579)
 121. Gonçalves T, Rio-Torto I, Teixeira LF, Cardoso JS (2022) A survey on attention mechanisms for medical applications: are we moving toward better algorithms? *IEEE Access* 10:98909–98935
 122. Kaur N, Mittal A (2022) CADxReport: chest X-ray report generation using co-attention mechanism and reinforcement learning. *Comput Biol Med* 145:105498
 123. Medjahed SA (2015) A comparative study of feature extraction methods in images classification. *Int J Image Graph Signal Process* 7(3):16
 124. Liu X (2022) Zhao C (2022) Research on image feature extraction algorithm of the egg and egg white protein thermal gelation based on PCA/ICA. *Comput Intell Neurosci* 1:1266332
 125. Salau AO, Jain S (2019) Feature extraction: a survey of the types, techniques, applications. In: 2019 International Conference on Signal Processing and Communication (ICSC), pp 158–164. IEEE
 126. Peng X, Zhang X, Li Y, Liu B (2020) Research on image feature extraction and retrieval algorithms based on convolutional neural network. *J Vis Commun Image Represent* 69:102705
 127. Chauhan K, Tomar H, Kamal K, Goel P (2023) Feature extraction from image sensing (remote): image segmentation. In: 2023 5th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N), pp 227–232. IEEE

128. Loukil Z, Mirza QKA, Sayers W, Awan I (2023) A deep learning based scalable and adaptive feature extraction framework for medical images. *Inf Syst Front* 1–27
129. Alzubaidi L, Zhang J, Humaidi AJ, Al-Dujaili A, Duan Y, Al-Shamma O, Santamaría J, Fadhel MA, Al-Amidie M, Farhan L (2021) Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. *J Big Data* 8:1–74
130. Baruah A, Saikia LP (2020) Study and analysis of different feature extraction methods in digital image processing. *Int J Comput Sci Mob Comput* 9:27–39

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